This standard has been approved by the Naval Ship Engineering Center, Department of the Navy and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. SHIPS
Other Cust

PROCUREMENT SPECIFICATION (B) FF-S-111

SUPERSEDES:

TITLE

SCREWS, WOOD, SLOTTED, FLAT COUNTERSUNK (82°) HEAD, COPPER-SILICON ALLOY, NONMAGNETIC

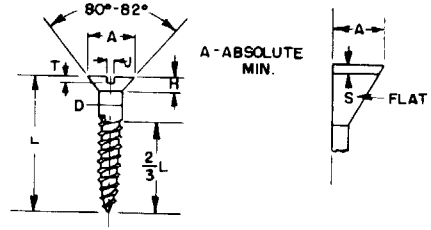
MILITARY STANDARD

MS16199(SHIPS)

SHEET 1 OF 2

FED. SUP CLASS 5305

INCH-POUND 5305-002231



NOMINAL SIZE		2	3	4	5	6	7	8	9	10	12	14	16	18	20	24	30
THREADS PER INCH		26	24	22	20	18	16	15	14	13	11	10	9	8	8	7	6
(D) BASIC DIAMETER	+0.004 -0.007	0.086	0.099	0.112	0.125	0.138	0.151	0.164	0.177	0.190	0.216	0.242	0.268	0.294	0.320	0.372	0.450
(A) HEAD DIAMETER	MAX. SHARP	.172	.199	.225	.252	.279	.305	.332	.358	.385	.438	.491	.544	.597	.650	.756	.915
	MIN. SHARP	.156	.181	.207	.232	.257	.283	.308	.334	.359	.410	.461	.512	.563	.614	.716	.882
	ABSOLUTE MIN. WITH FLAT	.147	.171	.195	.220	.244	.268	.292	.316	.340	.389	.437	.485	.534	.582	.679	.869
(H) HEIGHT OF HEAD	MAX.	.051	.059	.067	.075	.083	.091	.100	.108	.116	.132	.148	.164	.180	.196	.228	.276
	MIN.	.040	.048	.055	.062	.069	.076	.084	.091	.098	.112	.127	.141	.155	.170	.198	.228
(J) WIDTH OF SLOT	MAX.	.031	.035	.039	.043	.048	.048	.054	.054	.060	.067	.075	.075	.084	.084	.094	.102
	MIN.	.023	.027	.031	.035	.039	.039	.045	.045	.050	.056	.064	.064	.072	.072	.081	.084
(T) DEPTH OF SLOT	MAX.	.023	.027	.030	.034	.038	.041	.045	.049	.053	.060	.068	.075	.083	.090	.105	.128
	MIN.	.015	.017	.020	.022	.024	.027	.029	.032	.034	.039	.044	.049	.054	.059	.069	.084
(S) FLAT ON MIN. SCREW		.003	.004	.004	.005	.005	.005	.006	.006	.007	.008	.009	.010	.011	.012	.013	.015

NOTES

1. MATERIAL: MATERIAL SHALL BE COPPER-SILICON ALLOY IN ACCORDANCE WITH QQ-C-591, COPPER ALLOY NO. 651.
2. TENSILE STRENGTH: MINIMUM TENSILE STRENGTH SHALL BE 50,000 P.S.I.
3. DIMENSIONS: ALL DIMENSIONS ARE IN INCHES, UNLESS OTHERWISE SPECIFIED.
4. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS.
5. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 8 MAY 1957 REVISED (A) 8 MARCH 1963 REVISED (B) 8 MARCH 1967

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P.A. SHIPS
Other Cust

PROCUREMENT SPECIFICATION (B) FF-S-111

SUPERSEDES

TITLE

SCREWS, WOOD, SLOTTED, FLAT COUNTERSUNK (82°) HEAD, COPPER-SILICON ALLOY, NONMAGNETIC

MILITARY STANDARD

MS16199(SHIPS)

SHEET 2

SIZE NO.		2	3	4	5	6	7
LENGTH	LENGTH TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1/4	+0 -1/32	1	6	11	--	--	--
3/8		2	7	12	18	24	--
1/2		3	8	13	19	25	32
5/8		4	9	14	20	26	33
3/4		5	10	15	21	27	34
7/8		-	--	16	22	28	35
1	+0 -3/64	-	--	17	23	29	36
1-1/4		-	--	--	--	30	37
1-1/2		-	--	--	--	31	38

SIZE NO.		8	9	10	12	14	16
LENGTH	LENGTH TOLERANCE	DASH NO.	DASH NO	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1/2	+0 -1/32	39	--	126	--	--	--
5/8		40	48	56	--	--	--
3/4		41	**49**	57	66	--	--
7/8		42	**50**	58	67	--	--
1	+0 -3/64	43	51	59	68	78	127
1-1/4		44	52	60	69	79	128
1-1/2		45	53	61	70	80	89
1-3/4		46	54	62	71	81	90
2	+0 -1/16	47	55	63	72	82	91
2-1/4		--	--	64	73	83	92
2-1/2		--	--	65	74	84	93
2-3/4		--	--	--	75	85	94
3		--	--	--	76	86	95
3-1/2	+0 -3/32	--	--	--	77	87	96
4		--	--	--	--	88	97
4-1/2		--	--	--	--	--	--

SIZE NO.		18	20	24	30
LENGTH	LENGTH TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1-1/2	+0 -3/64	98	---	---	
2		99	107	---	
2-1/4	+0 -1/16	100	108	---	
2-1/2		101	109	---	
2-3/4		102	110	---	
3		103	111	115	----
3-1/2		104	112	116	----
4	+0 -3/32	105	113	117	1/ 121
4-1/2		106	114	118	1/ 122
5		---	---	---	1/ 129
5-1/2		---	---	1/ 119	1/ 123
6		---	---	1/ 120	1/ 124
7		---	---	---	1/ 125

1/ THESE ARE EXTRA LARGE SCREWS USED IN NONMAGNETIC MINESWEEPER APPLICATIONS.

FED SUP CLASS 5305

APPROVED 8 MAY 1957 REVISED (S) FOR CHANGES SEE SHEETS 1 AND 2.

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